

RATUL HASAN

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PROFESSIONAL SUMMARY

Computer Science and Engineering graduate from BRAC University and current departmental Research Assistant under Dr. Md. Golam Rabiul Alam. Experience in robotics, computer vision, embedded systems, and applied AI. Work ranges from robot hardware and locomotion to computer vision, edge AI assistants, and production mobile and web software. Two journal manuscripts are in preparation for ESWA and EAAI, focused on elderlycare vision and agentic assistance.

RESEARCH INTERESTS

Robotics and Autonomous Systems · Computer Vision · Machine Learning · Embedded AI · Human-Robot Interaction · Edge AI

EDUCATION

BRAC University <i>Bachelor of Science in Computer Science and Engineering</i>	Dhaka, Bangladesh 2022 – 2026
Govt. Bangabandhu College <i>Higher Secondary Certificate (Science), GPA: 5.00 / 5.00</i>	Bangladesh 2020
S.M. Model Govt. High School <i>Secondary School Certificate (Science), GPA: 5.00 / 5.00</i>	Bangladesh 2017

RESEARCH PAPERS

- “KAGE: A Knowledge-Guided Edge Expert System for Governed Agentic Assistance in Elderlycare.” **In preparation.** Target: *Expert Systems with Applications* (ESWA).
- “Feature-Cached Temporal Modeling for Multi-Category Behavioral Anomaly Detection in ElderlyCare Video Monitoring.” **In preparation.** Target: *Engineering Applications of Artificial Intelligence* (EAAI).
- Presentation:** Research poster on Advanced Driver Assistance Systems (ADAS) for future mobility.

RESEARCH & PROFESSIONAL EXPERIENCE

Research Assistant <i>Department of Computer Science and Engineering, BRAC University</i> - Departmental Research Assistant under the supervision of Dr. Md. Golam Rabiul Alam. - Conducting research in computer vision, edge AI, and agentic systems, including elderlycare video monitoring and knowledge-guided assistance	2026 – Present <i>Dhaka, Bangladesh</i>
Research Team Lead <i>BIOSE, Biomedical Science and Engineering Research Center, BRAC University</i> - Led student research on a sensor-driven bionic arm for prosthetic applications, focusing on Arduino-based actuation, servo control, and sensory feedback. - Designed and evaluated real-time finger movement and feedback mechanisms for modular robotic prosthetic systems. Recognized with an International Gold Award.	2024 – 2026 <i>Dhaka, Bangladesh</i>
Founder & CEO <i>Onnesok</i> - Founded and lead a tech venture shipping software and gaming products used by 100K+ users. Awarded UIHP Pre-seed Grant through the University Innovation Hub Program. - Developed Flutter and web applications with scalable backend services, real-time features, and AI-assisted academic tools.	2020 – Present <i>Dhaka, Bangladesh</i>
Mechanical & 3D Design Member <i>BRACU Duburi, BRAC University</i> - Member of BRACU Duburi, Bangladesh’s pioneering student-led Autonomous Underwater Vehicle (AUV) team that competes internationally in RoboSub. - Worked on the Mechanical and 3D Design sub-team designing AUV structural components and payload mechanisms (torpedo and dropper systems) for autonomous underwater missions.	2022 – 2023 <i>Dhaka, Bangladesh</i>

- Conduct hands-on training in Flutter and software engineering. Mentor students on architecture, clean code, and full-stack mobile development.

SELECTED PROJECTS

Robotics & Embedded Systems

- **Axelia:** Quadruped autonomous robot for terrain-adaptive locomotion, turning, and obstacle negotiation using modular mechanics and embedded motion control.
- **Sensory Feedback Bionic Arm:** Functional prosthetic prototype with sensor input, servo actuation, and sensory feedback for naturalistic finger movement. International Gold Award.
- **Pixi:** Interactive human-sized robotic platform with sensor-based feedback, real-time control, and modular hardware-software integration.
- **FarmBot:** Autonomous farming robot for planting, watering, and crop monitoring with hardware-software integration. National Winner, 10 Minute School Science Fair.

AI & Computer Vision Research

- **KAGE:** Knowledge-guided edge expert system for governed agentic assistance in elderlycare (in preparation for ESWA).
- **ElderlyCare Video Monitoring:** Feature-cached temporal modeling for multi-category behavioral anomaly detection (in preparation for EAAI).
- **ADECS:** Autonomous Anomaly Detection and Elderly Care System with robotic support, CCTV monitoring, and multi-sensor fusion.
- **AURA:** Agentic AI mobile assistant with long-term memory, task automation, and local LLM inference.

Software Systems & Products

- **TexLab Online:** Cloud-based LaTeX editor with real-time compilation and collaborative academic writing workflows (React, Next.js).
- **ShooterX:** Endless arcade shooter (Flutter, Flame) with power-ups, game modes, and global leaderboards. On Google Play and shooterx.space. BDApps Innovation Summit recognition (Play Store).
- **Adventure Of Sui:** Pixel-art mobile adventure on Google Play with offline play, leaderboards, and progression (Play Store).

AWARDS & HONORS

- **Vice Chancellor's Award:** BRAC University, Excellence in Extracurricular Activities.
- **UIHP Pre-seed Grant:** University Innovation Hub Program funding for Onnesok.
- **International Gold Award:** Sensory Feedback Bionic Arm.
- **IEEE YESIST12:** National Winner and International Competitor.
- **Technoxian World Robotics Challenge:** National Champion.
- **IoT Matrix Hackathon:** National Winner.
- **10 Minute School Science Fair:** National Winner (FarmBot).
- **BDApps Innovation Summit:** 2nd Runner-Up (ShooterX).
- **NHSPC:** Regional Winner, Competitive Programming.

TECHNICAL SKILLS

Languages: Python, C/C++, Dart

AI & Vision: OpenCV, Computer Vision, Machine Learning, LLM Integration, Agentic AI Systems

Robotics & Embedded: Arduino, ESP32, Raspberry Pi, Sensors, Embedded Control, Real-Time Systems, IoT

Software & Systems: Flutter, React, Next.js, Node.js, Firebase, PostgreSQL, MongoDB, Docker, Linux

Tools: Git, Bash, GCP, Azure

LEADERSHIP

- **Secretary, ROBU** (Robotics Club of BRAC University), 2022–2026: Coordinated robotics projects, documentation, and club operations.
- **Student Developer, BRAC Skill Development Program (SDP)**, 2025–2026: Built an AI-assisted career platform for job discovery, candidate matching, and career counseling.
- **Facilitator, Ecollab International**, 2023–2024: Supported international climate collaboration across Bangladesh, Nigeria, Jordan, Indonesia, and Brazil.